ERCP Timing Before Deterioration in Acute Cholangitis Using Bilirubin WBC Creatinine Platelets and Albumin Through a Stochastic Optimal Stopping Model

Prihantini
Master Student of Financial Engineering, WorldQuant University, Washington. D.C, USA

Abstract

Acute cholangitis is a time-sensitive consequence of biliary obstruction, but the exact ERCP window remains clinically nuanced. Guidelines support drainage and commonly use a 48-hour boundary, observational syntheses favor earlier drainage, while a 2026 randomized trial in mild-to-moderate disease found no mortality or organ-failure advantage for <24-hour ERCP over 24-48 hours and reported more procedure-related adverse events with urgent ERCP. We analyzed a literature-parameterized mathematical proof-of-concept that treats ERCP timing as a stochastic optimal-stopping problem using five routinely available variables bilirubin, creatinine, platelets, and albumin to represent cholestatic burden, inflammation, organ dysfunction, and physiologic reserve. The model combined five literature-normalization functions, an eight-state continuous dynamical system for the five laboratory variables plus inflammatory burden I(t), organ dysfunction O(t), and stability reserve R(t), a four-state continuous-time Markov regime model, competing waiting and procedure intensities, and a 72-hour finite-horizon Bellman optimal-stopping calculation. Numerical work used adaptive RK45, Latin-hypercube sampling with PRCC, ±1% local elasticity, stochastic input perturbation. At 24 hours the simulated deterioration index was 0.471 and stability reserve 0.220. Stochastic perturbations yielded a median stopping time of 28 hours with a 2.5^{th}-97.5^{th} percentile range of 19-40 hours. All eight post-drainage eigenvalues were negative (-0.180 to -0.035), supporting local asymptotic stability of the recovery equilibrium in the specified subsystem. The three largest absolute PRCCs were hwait0 -0.920, wait_beta -0.910, hproc0 0.802. These are mathematical outputs and the current model establishes a mathematically testable structure for a familiar bedside problem. When continued stabilization becomes less favorable than biliary drainage as laboratory evidence of deterioration accumulates. The main advantage is that optimal stopping links time, evolving physiology, and competing procedure versus waiting costs in a way that static association models do not. The model can support retrospective target-trial emulation and eventual prospective decision.

***Keywords*:** *Acute Cholangitis, Clinical Deterioration, ERCP, Optimal Stopping, Stochastic Decision Model, Uncertainty Quantification*

1. Background

Acute cholangitis is a potentially life-threatening biliary infection in which obstruction, systemic inflammation, and organ dysfunction may require urgent intervention. Although antimicrobial therapy and initial resuscitation are essential, biliary drainage remains a central component of definitive management. The Tokyo Guidelines 2018 (TG18) provide a structured approach to severity assessment and treatment decisions, distinguishing moderate disease through combinations of inflammatory and biochemical abnormalities and severe disease through evidence of organ dysfunction. White blood cell count, total bilirubin, serum albumin, creatinine, and platelet count are particularly relevant to these assessments, although they represent different components of the grading criteria and do not independently establish overall clinical severity (Kiriyama et al., 2018 and Miura et al., 2018). The practical challenge is not simply identifying patients who require drainage, but determining how changes in physiological condition during initial treatment should influence the timing of endoscopic retrograde cholangiopancreatography (ERCP).

The evidence supporting early biliary drainage is substantial, but the appropriate timing within the first 48 hours remains dependent on clinical severity and procedural considerations. A systematic review and meta-analysis involving 84,063 patients found that ERCP performed within 48 hours was associated with lower in hospital mortality than later intervention (odds ratio 0.57, 95% confidence interval 0.51-0.63), although the included studies were observational (Du et al., 2020). Consistent with these findings, the American Society for Gastrointestinal Endoscopy recommends biliary decompression within 48 hours, while recognizing the conditional nature and very low certainty of the supporting evidence (Buxbaum et al., 2021). More recently, a randomized trial involving 304 patients with mild to moderate acute cholangitis found no statistically significant difference in 30-day mortality or organ failure between ERCP performed within 24 hours and intervention at 24-48 hours. Procedure related adverse events were more frequent in the urgent intervention group, although these findings should not be extrapolated to patients with severe cholangitis or septic shock (Jagtap et al., 2026). These studies support timely drainage while indicating that earlier intervention is not necessarily associated with uniformly better outcomes in every severity group.

A mathematical difficulty arises because conventional severity classifications and comparisons between predefined intervention intervals do not directly represent how the consequences of waiting may change with evolving physiology. A severity grade summarizes clinical condition at an assessment point, whereas the decision to continue stabilization or proceed with drainage is inherently sequential. In mathematical terms, this distinction motivates a dynamic decision problem in which physiological variables evolve over time and the relative costs of intervention and continued waiting depend on the simulated state. Optimal stopping theory provides a formal basis for comparing immediate action with continuation over a finite horizon, while free boundary concepts describe the separation between regions in which stopping or waiting is favored (Peskir and Shiryaev, 2006). The application is methodological and the objective is to examine a clinical timing decision using mathematical model to develope for time dependent decisions under uncertainty. Importantly, the present implementation evaluates stopping decisions along simulated physiological trajectories. It does not estimate a fully patient-calibrated stochastic decision policy.

The present study develops a literature-parameterized mathematical proof-of-concept that connects TG18 informed indicators with nonlinear physiological dynamics and a finite horizon stopping calculation. An eight-state ordinary differential equation system represents bilirubin, white blood cell count, creatinine, platelets, albumin, inflammatory burden, organ dysfunction, and physiological reserve.

A four-state continuous time model provides a complementary characterization of possible clinical regimes, while competing waiting and procedure related intensities enter the decision cost formulation. The resulting model is examined through prespecified laboratory scenarios, Latin hypercube sampling, partial rank correlation analysis, and stochastic input perturbations. The primary objective is to determine how assumed physiological evolution and competing decision costs influence candidate ERCP stopping times and their uncertainty. A secondary objective is to identify which model parameters most strongly determine these times and which assumptions require empirical calibration. By connecting clinical severity indicators with dynamic decision analysis, this study establishes a reproducible theoretical basis for subsequent patient level investigation, without interpreting simulated stopping times as validated treatment thresholds.

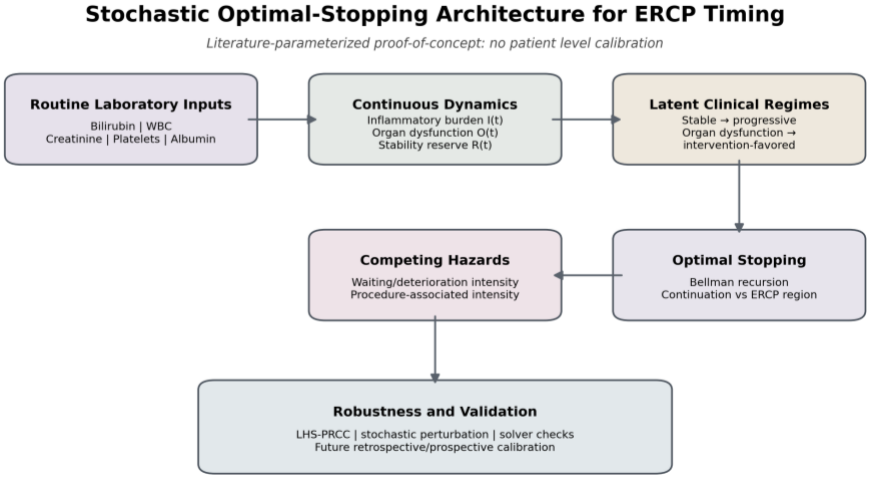


Figure 1. *Hybrid architecture linking routine laboratory inputs to disease dynamics, latent regimes, competing hazards, and the optimal-stopping layer.*

2. Methods

2.1. Study Design

This study developed a literature-parameterized nonlinear mathematical model to investigate the timing of endoscopic retrograde cholangiopancreatography (ERCP) in acute cholangitis under evolving biochemical abnormalities, inflammatory burden, and organ dysfunction. The computational approach combined an eight-dimensional ordinary differential equation (ODE) system, a four-state continuous time Markov chain (CTMC), competing deterioration and procedure-related cost intensities, and finite-horizon optimal stopping. The study was conducted as a deterministic and stochastic mathematical proof-of-concept. No patient level records, individual treatment outcomes, or longitudinal laboratory measurements were used for parameter estimation, model fitting, or clinical validation. The numerical trajectories represent simulated responses build from explicitly specified initial conditions and provisional model coefficients.

Literature was obtained from the Tokyo Guidelines 2018 (TG18), including diagnostic and severity grading criteria, initial management recommendations, and biliary drainage principles (Kiriyama et al., 2018, Miura et al., 2018, Mukai et al., 2018 and Mayumi et al., 2018). The American Society for Gastrointestinal Endoscopy (ASGE) guideline provided an additional reference for biliary decompression timing (Buxbaum et al., 2021). Evidence concerning the association between ERCP timing and clinical outcomes was considered from published systematic reviews and meta-analyses (Du et al., 2020; Lyu et

al., 2022). The evidence was incorporated at three distinct levels. First, published clinical criteria defined laboratory reference thresholds and severity related interpretations. Second, mathematical assumptions specified the direction and strength of interactions among laboratory markers, inflammatory activity, organ dysfunction, and physiological reserve. Third, numerical simulations evaluated how these assumed interactions affected theoretical intervention timing.

Published laboratory thresholds were not treated as estimates of continuous disease progression rates. Likewise, associations reported in clinical studies were not directly converted into ODE coefficients, transition intensities, or intervention cost functions. Parameters without an identifiable empirical estimate were classified as provisional structural or decision coefficients and evaluated through uncertainty analysis. The primary computational endpoint was the model based first stopping hour, defined as the earliest hourly decision point at which the theoretical cost of immediate ERCP did not exceed the modelled cost of continued observation. Secondary endpoints included laboratory trajectories, the composite deterioration index, stability reserve, the bilirubin creatinine stopping surface, local equilibrium stability, and the distribution of stopping times under parameter and initial-condition perturbations. The model was intended to examine mathematical decision behaviour under transparent assumptions, not to establish an independently validated clinical ERCP scheduling rule.

2.2. Literature Parameters

Parameters were selected from TG18 severity criteria (Kiriyama et al., 2018). ERCP timing recommendations were used for contextual interpretation, not imposed as fixed stopping constraints (Buxbaum et al., 2021, Du et al., 2020 and Lyu et al., 2022).

Table 1. Literature Parameters

Domain	Anchor	Role / source
Bilirubin	≥5 mg/dL	TG18 Grade II (Kiriyama et al., 2018)
WBC	>12 or <4 ×10³/μL	TG18 Grade II (Kiriyama et al., 2018)
Creatinine	>2 mg/dL	TG18 Grade III (Kiriyama et al., 2018)
Platelets	<100 ×10³/μL	TG18 Grade III (Kiriyama et al., 2018)
Albumin	<0.70 × LLN	TG18 Grade II (Kiriyama et al., 2018)
Biliary decompression	Within 48 h	Conditional ASGE guidance (Buxbaum et al., 2021)
ERCP timing	24/48/72-hour comparisons	Clinical context (Du et al., 2020 and Lyu et al., 2022)

The albumin logistic center of 2.8 g/dL assumes an illustrative institutional lower limit of normal of 4.0 g/dL. This is a modeling convention, not a universal clinical threshold. The continuous transformations do not reproduce the complete TG18 severity-grading algorithm.

2.3. State Variables and Initial Conditions

Table 2. Variables

Symbol	State	Unit / range	Reference initial value
$\boldsymbol{B(t)}$	Total bilirubin	mg/dL	6.00
$\boldsymbol{W(t)}$	White blood cells	$10^3/\mu L$	14.00
$\boldsymbol{C(t)}$	Creatinine	mg/dL	1.35
$\boldsymbol{P(t)}$	Platelets	$10^3/\mu L$	145.00
$\boldsymbol{A(t)}$	Albumin	g/dL	3.05
$\boldsymbol{I(t)}$	Inflammatory burden	0–1 latent	0.22
$\boldsymbol{O(t)}$	Organ dysfunction	0–1 latent	0.10
$\boldsymbol{R(t)}$	Stability reserve	0–1 latent	0.80

The five laboratory inputs and three latent states were initialized at illustrative reference values, not reconstructed from a patient record. The latent states are dimensionless mathematical constructs, not validated clinical scores. Time t was measured in hours.

2.4. Mathematical Formulation

2.4.1. Continuous Models

The values were converted into dimensionless quantities using logistic functions. The logistic scale coefficients are provisional numerical choices rather than fitted clinical estimates

$$\sigma(x) = \frac{1}{[1 + \exp(-x)]}$$

$$b(B) = \frac{1}{\left[1 + \exp\left(-\frac{B - 5.0}{1.4}\right)\right]}$$

$$w(W) = \max\left\{\frac{1}{\left[1 + \exp\left(-\frac{W - 12}{2.8}\right)\right]}, \frac{1}{\left[1 + \exp\left(-\frac{4 - W}{1.2}\right)\right]}\right\}$$

$$c(C) = \frac{1}{\left[1 + \exp\left(-\frac{C - 2.0}{0.45}\right)\right]}$$

$$p(P) = \frac{1}{\left[1 + \exp\left(-\frac{100 - P}{18}\right)\right]}$$

$$a(A) = \frac{1}{\left[1 + \exp\left(-\frac{2.8 - A}{0.30}\right)\right]}$$

The WBC transformation captures both leukocytosis and leukopenia, consistent with TG18 (Kiriyama et al., 2018).

2.4.2. Nonlinear Dynamical Model

$$\frac{dB}{dt} = k_B(1 + 0.55I) - k_B \max\left\{\frac{(B-2),0}{5}\right\}$$

$$\frac{dW}{dt} = -k_W \max\{(W-8),0\} + k_W B[3b(B) + 2I]$$

$$\frac{dC}{dt} = k_C(0.8I + 1.4O) - k_C \max\{((C-0.9),0)R\}$$

$$\frac{dP}{dt} = -k_P(8I + 10O) + k_P \max\left\{\frac{((180-P),0)R}{80}\right\}$$

$$\frac{dA}{dt} = -k_A(1.2I + 0.8O) + k_A \max\{((3.7 - A),0)R\}$$

$$\frac{dI}{dt} = [k_I Bb(B) + k_I Ww(W)](1-I) - k_I RRI$$

$$\frac{dO}{dt} = [k_O II + k_O Cc(C) + k_O Pp(P)](1-O) - k_O RRO$$

$$\frac{dR}{dt} = k_R + (1-R) - k_R IIR - k_R OOR - k_R Aa(A)R$$

Coefficients k denote provisional rates and couplings recorded in the project parameter tables. These equations encode assumed interactions and should not be interpreted as empirically established causal laws. Their units must be chosen to match each laboratory derivative per hour.

2.4.3. Composite Deterioration Index

$$D(t) = 0.16b(B) + 0.18w(W) + 0.22c(C) + 0.18p(P) + 0.10a(A) + 0.16O$$

The weights sum to one and were chosen for this proof-of-concept. They were not derived from patient level regression or expert consensus. $D(t)$ is a model specific dimensionless index.

2.4.4. Four state continuous-time Markov chain

Four modeled states were Stable (S0), Progressive (S1), Organ Dysfunction (S2), and Intervention-Favored (S3). Forward and selected reverse transitions were allowed. The generator was:

$$Q = [[-q01, q01, 0, 0], [q10, -(q10 + q12), q12, 0], [0, q21, -(q21 + q23), q23], [0, 0, q32, -q32]]$$

$$q01(t) = q01_{base} \times \exp(2.2D(t))$$

$$q12(t) = q12_{base} \times \exp(2.6 \max(D(t) - 0.18, 0))$$

$$q23(t) = q23_{base} \times \exp(3.0 \max(D(t) - 0.34, 0))$$

$$q10(t) = q10_{base} \times \exp(1.5 \max(R(t) - 0.55, 0))$$

$$q21(t) = q21_{base} \times \exp(\max(R(t) - 0.50, 0))$$

$$q32(t) = q32_{base}$$

$$\pi(0) = [0.780, 0.170, 0.045, 0.005]$$

$$\pi(t + \Delta t) = \pi(t) \times \exp(Q(t)\Delta t)$$

Transition multipliers and thresholds were assumed rather than estimated from clinical transition data. The model was held constant within each numerical propagation step. The CTMC illustrated regime occupancy and did not directly determine the Bellman stopping rule.

2.4.5. Waiting Simulation Model

$$\lambda_{wait}(t) = h_W 0 \times \exp\{\beta_W[0.70D(t) + 0.30O(t)]\}$$

$$\lambda_{ERCP}(t) = h_P 0\left[1 + \psi \exp\left(-\frac{t}{\tau}\right)\right] \times \exp\{\beta_P[0.55O(t) + 0.45(1 - R(t))]\}$$

Here $h_W 0$ and $h_P 0$ denote baseline theoretical intensities, β_W and β_P govern state dependence, ψ and τ describe the assumed early procedure penalty and its decay. These are cost intensities. The functional forms are mathematical assumptions motivated by the clinical importance of severity, stabilization, and drainage timing (Miura et al., 2018 and Buxbaum et al., 2021).

2.4.6. Finite Horizon Optimal Stopping

$$C_{stop}(t) = c_0 + \lambda_{ERCP}(t) + 0.018O(t)$$

$$C_{wait}(t) = c_1 + \lambda_{wait}(t) + 0.010D(t) + \gamma V(t + 1)$$

$$V(t) = \min[C_{stop}(t), C_{wait}(t)]$$

$$V(72) = c_T[0.55D(72) + 0.45O(72)]$$

$$\tau* = \min\{t \in \{0, 1, \ldots, 71\} : C_{stop}(t) \leq C_{wait}(t)\}$$

The finite horizon Bellman recursion was solved backward from 72 hours using hourly sampled trajectories. If no earlier stop was selected, the terminal hour was assigned. The resulting time is a trajectory conditioned theoretical stopping decision.

2.5. Theoretical Analysis

2.5.1. Existence, uniqueness, and state constraints

$$X(t) = [B(t), W(t), C(t), P(t), A(t), I(t), O(t), R(t)]$$

$$\frac{dX}{dt} = F(X, \theta)$$

Logistic functions are smooth and max operators are locally Lipschitz, giving local existence and uniqueness for finite parameters and states. For nonnegative relevant coefficients, boundary signs of the I, O, and R equations make [0,1] forward invariant. The laboratory equations do not establish global physiological positivity for all possible states, numerical clipping in the fast RK4 implementation was treated as a computational safeguard, not an analytical proof.

2.5.2. Post Drainage reference equilibrium and local stability

$$X* = [2.0, 8.0, 0.9, 180, 3.7, 0, 0, 1]$$

$$J_{post} = \frac{\partial F_{post}}{\partial X \; evaluated \; at \; X*}$$

$$det(J_{post} - \lambda I_8) = 0$$

$$\lambda = [-0.065, -0.085, -0.055, -0.045, -0.035, -0.180, -0.150, -0.050]$$

The project reports eight negative real eigenvalues. If these were calculated from the Jacobian of a correctly specified post-drainage vector field at an actual equilibrium, they imply local asymptotic stability. The stated X* is not an equilibrium of the untreated pre drainage system, so the result cannot be used to claim stability of that system. Global or Lyapunov stability was not established.

2.5.3 Probability conservation and identifiability

$$qij \geq 0 \; for \; i \neq j; \;\; \Sigma_j \, qij = 0, \;\; \Sigma_i \, \pi_i(t) = 1$$

The Markov generator has nonnegative off diagonal rates and zero row sums, ensuring probability conservation. Since no longitudinal outcomes were fitted, neither structural nor practical parameter identifiability was defined. Sensitivity analysis is not a substitute for clinical parameter identification.

2.6 Computational Analysis

2.6.1 Integration and stopping evaluation

The reference eight state ODE was integrated from 0 to 72 hours with adaptive RK45, output every 0.25 hours, relative tolerance 1×10^{-8}, and absolute tolerance 1×10^{-10}. Hourly states were passed to the backward Bellman recursion. The code also included a faster fixed step fourth order Runge-Kutta routine with a one-hour step and numerical state constraints for repeated simulations. The recorded random seed was 20260819.

2.6.2 Stopping surface, scenarios, and phase plane

The two-dimensional stopping surface varied initial bilirubin from 2 to 12 mg/dL and creatinine from 0.8 to 3.0 mg/dL, holding other reference inputs fixed. Eight illustrative scenarios changed laboratory severity and latent reserve. The I-O phase plane visualized the coupled inflammatory and organ dysfunction dynamics. Markov probabilities were propagated by the matrix exponential. No PDE, finite difference, or finite element solver was required.

$$T_{stop} = f(B(0), C(0) \mid \theta, X_{other}(0))$$

$$Trajectory(t) = [I(t), O(t)]$$

2.7 Robustness and Uncertainty Analysis

2.7.1 Latin hypercube sampling and PRCC

Global parameter uncertainty was evaluated with 600 Latin hypercube parameter vectors over the provisional bounds documented in the project. The sampled factors included inflammatory and organ dysfunction coupling, reserve dynamics, waiting and procedure intensities, early procedure penalty, and intensity slopes. The resulting stopping hours were analyzed using partial rank correlation coefficients (PRCC), with rank residualization against the remaining sampled parameters.

$$\theta_{i,j} = L_j + u_{i,j}(U_j - L_j)$$

$$PRCC_j = corr(residual_{rank}(\theta_j), residual_{rank}(T_{stop}))$$

PRCC signs describe modeled associations, not causal clinical effects. Computational p-values were not interpreted as evidence from patient observations.

2.7.2 Local elasticity

$$E_j \approx [T_{stop}(1.01\theta_j) - T_{stop}(0.99\theta_j)] / (0.02T_{ref})$$

Each selected parameter was perturbed by ±1% about the reference configuration. Because stopping hours are discrete, local responses can be zero or discontinuous; these values were not interpreted as smooth derivatives.

2.7.3 Stochastic initial-condition perturbations

A separate 1000-run experiment sampled the five laboratory inputs and three latent initial states around the reference condition. Normal draws were clipped to the specified intervals rather than resampled as exact truncated-normal distributions.

Variable	Sampling distribution (mean, SD)	Clipping limits
Bilirubin	Normal (6.0, 1.4)	2-12 mg/dL
WBC	Normal (14.0, 2.2)	7-22 ×10³/μL
Creatinine	Normal (1.35, 0.35)	0.7-3.0 mg/dL
Platelets	Normal (145, 25)	60-220 ×10³/μL
Albumin	Normal (3.05, 0.35)	1.8-4.2 g/dL
Inflammatory burden	Normal (0.22, 0.05)	0.05-0.50
Organ dysfunction	Normal (0.10, 0.05)	0-0.45
Stability reserve	Normal (0.80, 0.08)	0.35-0.98

Stopping times were summarized by the median and 2.5th-97.5th percentiles. These intervals characterize assumed computational uncertainty.

2.7.4 Numerical verification and statistical scope

Checks included solver completion, latent-state constraints, CTMC probability conservation, and stopping time robustness. AIC and BIC were not used because no likelihood was fitted to patient outcomes. The study evaluates numerical reproducibility and internal mathematical behavior, not clinical calibration, discrimination, or treatment effects.

3. Results

3.1 Reference Trajectory and Candidate Stopping Point

The reference profile was initialized at bilirubin 6.00 mg/dL, white blood cell count (WBC) 14.00 × 10³/μL, creatinine 1.35 mg/dL, platelets 145 × 10³/μL, albumin 3.05 g/dL, inflammatory burden $I(0) = 0.22$, organ dysfunction $O(0) = 0.10$, and stability reserve $R(0) = 0.80$. Under the modeled 72-hour no drainage trajectory, bilirubin and creatinine increased, while platelets and albumin decreased. WBC

decreased despite increasing latent inflammatory burden; therefore, the simulated laboratory markers did not all deteriorate monotonically (Figure 2).

At 24 hours, bilirubin was 6.26 mg/dL, WBC 12.05 × 10^3/μL, creatinine 1.65 mg/dL, platelets 143.1 × 10^3/μL, and albumin 2.95 g/dL. The composite deterioration index was $D(24) = 0.471$ and stability reserve was $R(24) = 0.220$. In the latent subsystem, inflammatory burden and organ dysfunction rose toward high values while reserve fell toward approximately 0.20. All values are numerical outputs, not patient measurements.

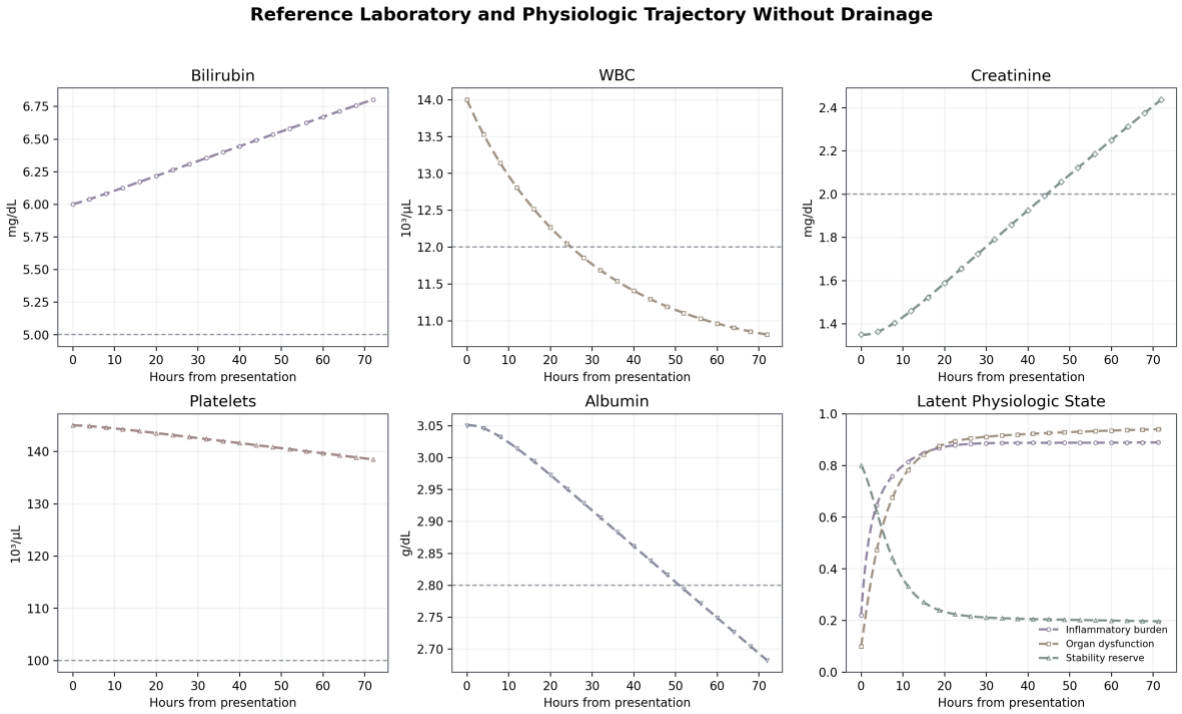


Figure 2. *Plot latent state trajectory under continued observation without modelled drainage. Dashed lines are literature anchors.*

3.2 Competing Intensities and Finite Horizon Stopping Decision

The waiting related intensity increased over the simulated horizon, whereas the procedure associated intensity decreased from its initially elevated value (Figure 3). These curves represent assumed theoretical intensity functions. Backward Bellman evaluation of the hourly trajectory identified the first drainage favored decision at 29 hours (Figure 3). The decision arose from comparison of the modeled immediate drainage and continuation costs. It was not defined by the intersection of the two intensity curves alone.

Before the first stopping hour, the continuation value was lower than the immediate drainage value. At approximately 29 hours the ordering reversed, and immediate drainage became the lower cost action within the model. The plotted terminal hour behavior reflects the finite horizon boundary condition and should not be interpreted as a biological discontinuity.

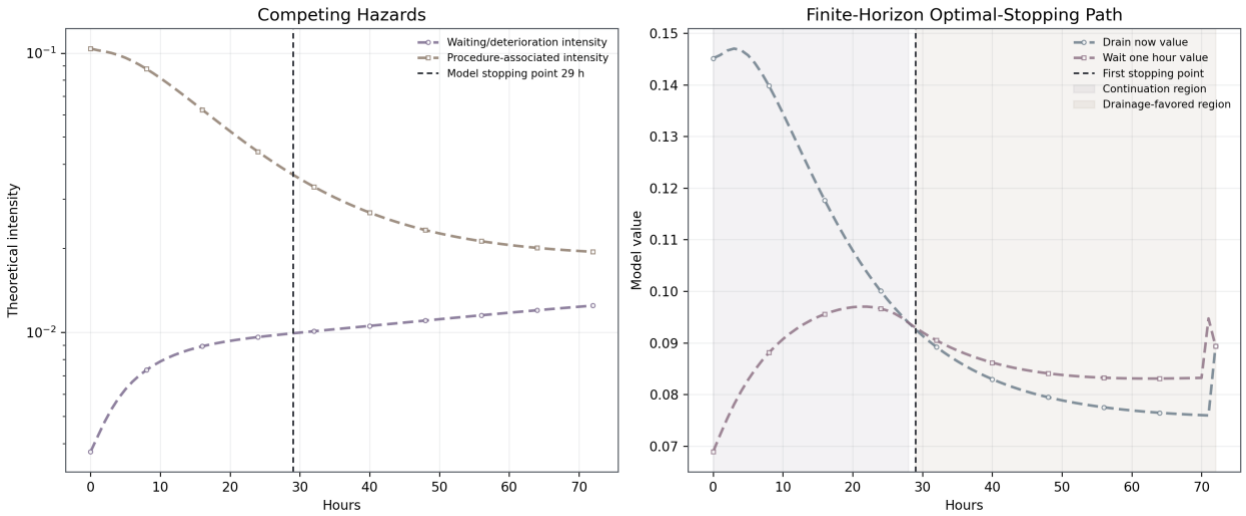


Figure 3. *Competing theoretical waiting and procedure-related intensities (left), and finite horizon Bellman decision values (right). The dashed vertical line denotes the reference stopping decision at 29 hours, shaded regions identify the modeled continuation and drainage favored intervals.*

3.3 Scenario Specific Shifts in Candidate Intervention Timing

The eight prespecified scenarios produced stopping decisions between 8 and 57 hours (Table 3 and Figure 4). Relative to the reference stopping hour of 29, the combined severe laboratory profile advanced the decision to 8 hours, thrombocytopenic deterioration and emerging renal dysfunction produced stopping hours of 16 and 21, respectively. Marked cholestasis and hypoalbuminemic low reserve each produced 25 hours. In contrast, the borderline moderate profile and lower inflammatory burden yielded 35 and 57 hours, respectively. The corresponding 24-hour deterioration indices ranged from 0.272 to 0.839, while stability reserve ranged from 0.181 to 0.289. These results show variation under the predefined scenario inputs and model cost structure, rather than observed differences in treatment benefit. The shaded 24-48-hour interval in Figure 4 is included only as a literature informed timing context; the 57-hour result is not a recommendation to delay indicated drainage.

Table 3. Scenario Outputs from the Proof-of-Concept Model

Scenario	Stopping hour	D(24 h)	R(24 h)
Reference moderate profile	29	0.471	0.220
Lower inflammatory burden	57	0.272	0.289
Marked cholestasis	25	0.520	0.217
Renal dysfunction emerging	21	0.584	0.209
Thrombocytopenic deterioration	16	0.675	0.202
Hypoalbuminemic low reserve	25	0.514	0.191
Combined severe laboratory profile	8	0.839	0.181
Borderline moderate profile	35	0.398	0.233

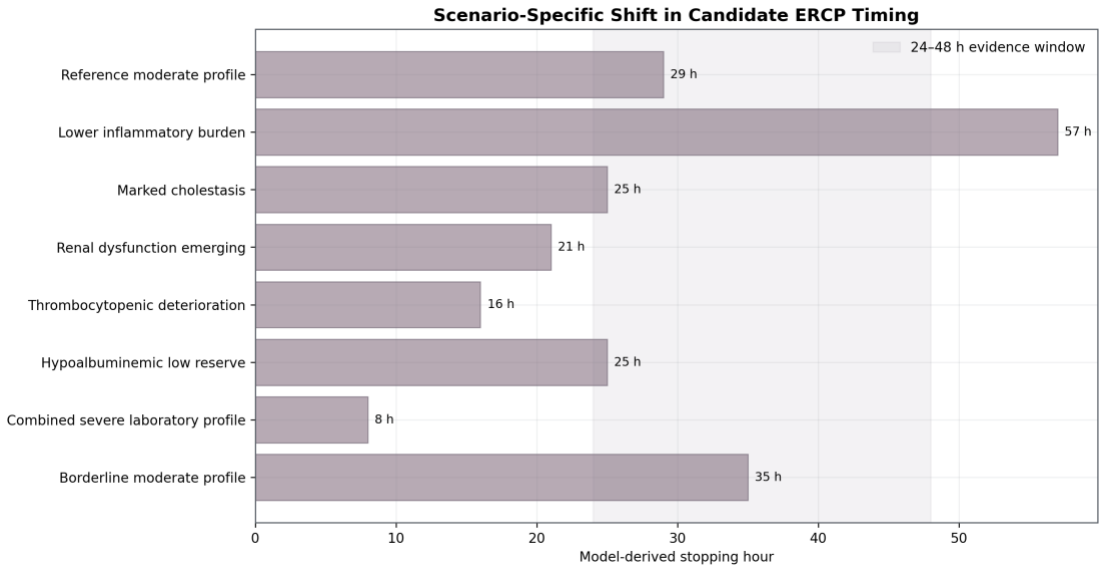


Figure 4. *Candidate ERCP stopping hours across eight illustrative profiles. Scenario-specific candidate ERCP timing shifts with the 24-48-hour window shown for context.*

3.4 Bilirubin Creatinine Stopping Surface

The two-dimensional input sweep evaluated 483 initial bilirubin creatinine combinations, spanning bilirubin 2-12 mg/dL and creatinine 0.8-3.0 mg/dL while holding other inputs at their reference values. The model derived stopping hours ranged from 15 to 52 (Figure 5). Higher baseline bilirubin and creatinine generally corresponded to earlier stopping decisions. The reference point, bilirubin 6.0 mg/dL and creatinine 1.35 mg/dL, lay near the 29-hour contour. The stepped contour boundaries arise in part from the one-hour decision grid. The surface is a discrete numerical decision map, empirically estimated clinical threshold or a formally solved continuous time free boundary.

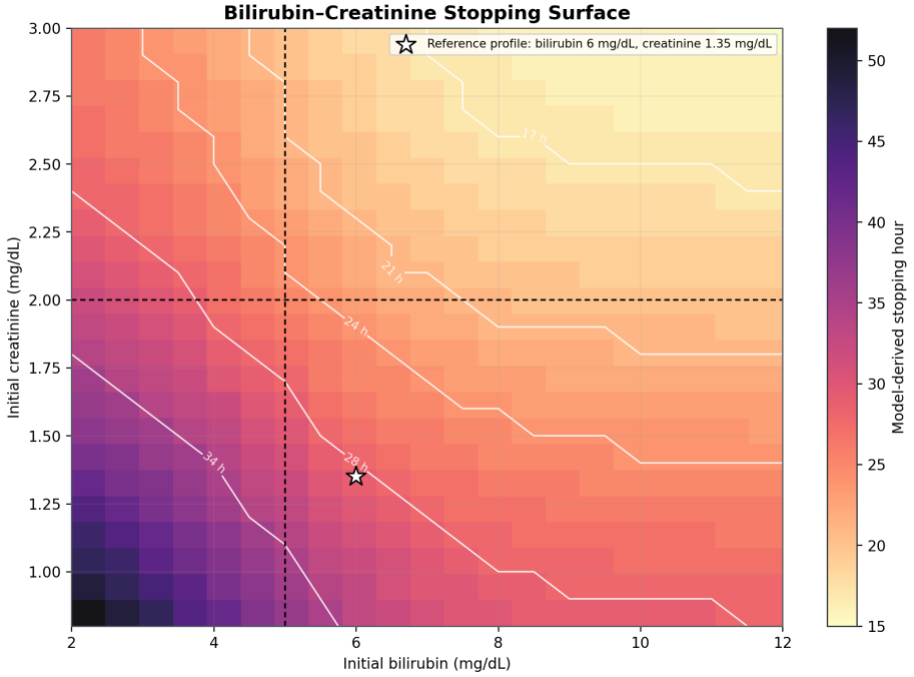


Figure 5. *Numerical stopping hour surface across 483 bilirubin creatinine starting profiles. The star identifies the reference initial profile, dashed guides indicate selected laboratory reference values. Contours represent integer hour model decisions.*

3.5 Latent State Phase Plane Behavior

The inflammatory burden organ dysfunction phase plane trajectory followed a curved, nonproportional path (Figure 6). Starting from $I(0) = 0.22$ and $O(0) = 0.10$, both latent variables increased during continued observation and approached high modeled burdens by the end of the horizon. The color progression indicates elapsed simulation time. This visualization describes the interaction encoded in the nonlinear equations; it does not show a clinically observed sequence of organ injury.

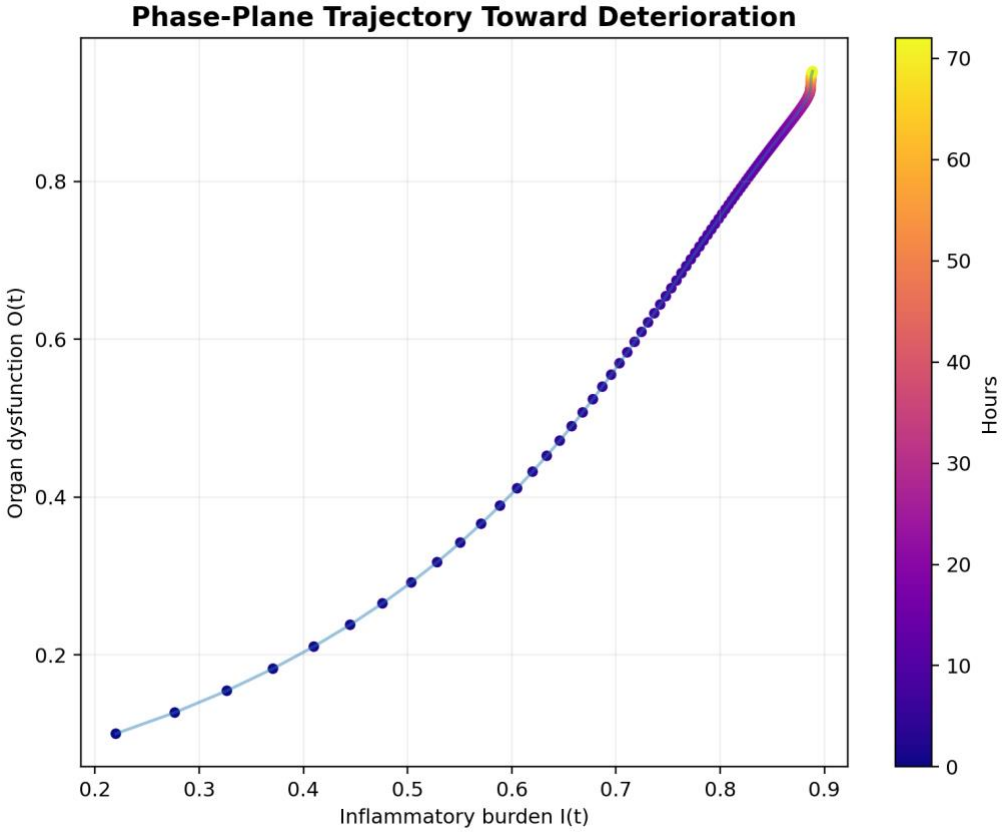


Figure 6. *Phase plane trajectory of latent inflammatory burden I(t) and organ dysfunction O(t) over 72 hours. Color indicates simulated time.*

3.6 Markov Regime Occupancy and Theoretical Post Drainage Stability

The four state CTMC began with probabilities of 0.780 stable, 0.170 progressive, 0.045 organ dysfunction, and 0.005 intervention favored. Over continued observation, stable state occupancy declined while the intervention favored probability increased (Figure 7). Near the 29-hour stopping decision, the stable state probability was approximately 0.30, with substantial probability in progressive and organ dysfunction regimes. These are model state occupancies and do not represent estimated patient level transition frequencies. The separate post drainage reference analysis reported eight real eigenvalues: -0.180, -0.150, -0.085, -0.065, -0.055, -0.050, -0.045, and -0.035 h^{-1}, with zero imaginary components. All are negative, consistent with local asymptotic stability of the stated linearized recovery system if the reference state is an equilibrium of its explicitly defined post-drainage equations. This result does not establish stability of the untreated trajectory, global convergence, or clinical recovery after ERCP.

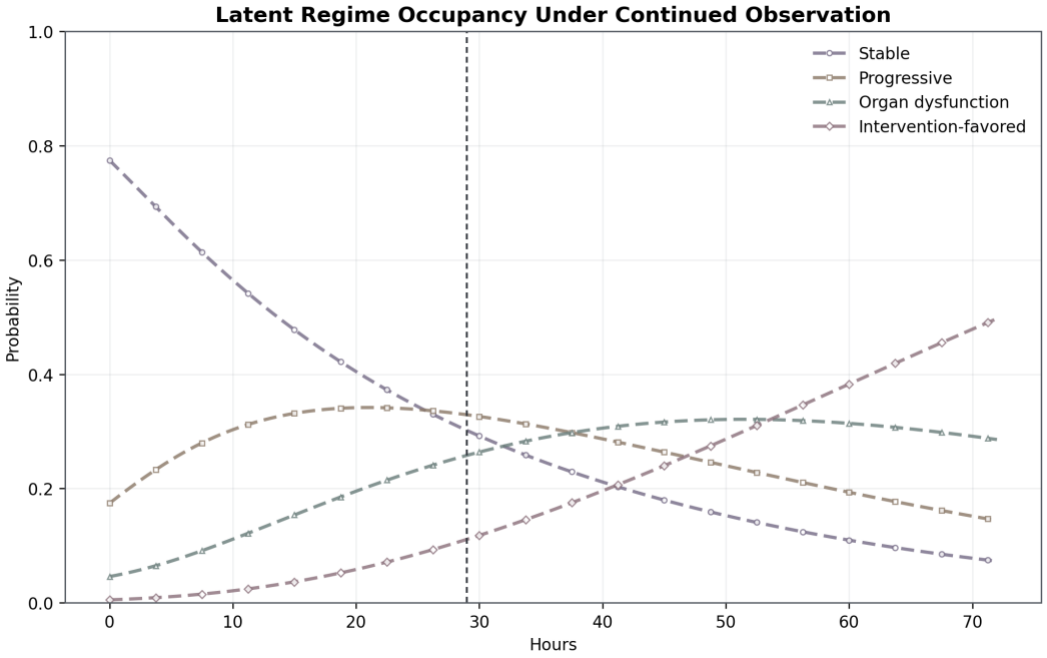


Figure 7. *CTMC probabilities for stable, progressive, organ dysfunction, and intervention-favored states during continued observation. The dashed line marks the reference 29-hour decision. Occupancy is independent of the Bellman decision calculation.*

3.7 Global Parameter Sensitivity and Local Elasticity

Across 600 Latin hypercube parameter sets, stopping time was most strongly associated with the decision-intensity coefficients (Figure 8). Waiting baseline intensity (hwait0, PRCC = -0.920) and waiting slope (wait_beta, -0.910) were associated with earlier stopping, while procedure baseline intensity (hproc0, 0.802), early procedure penalty (0.728), procedure decay (0.703), and procedure slope (0.531) were associated with later stopping. Physiological coupling coefficisents showed substantially smaller absolute PRCC values.

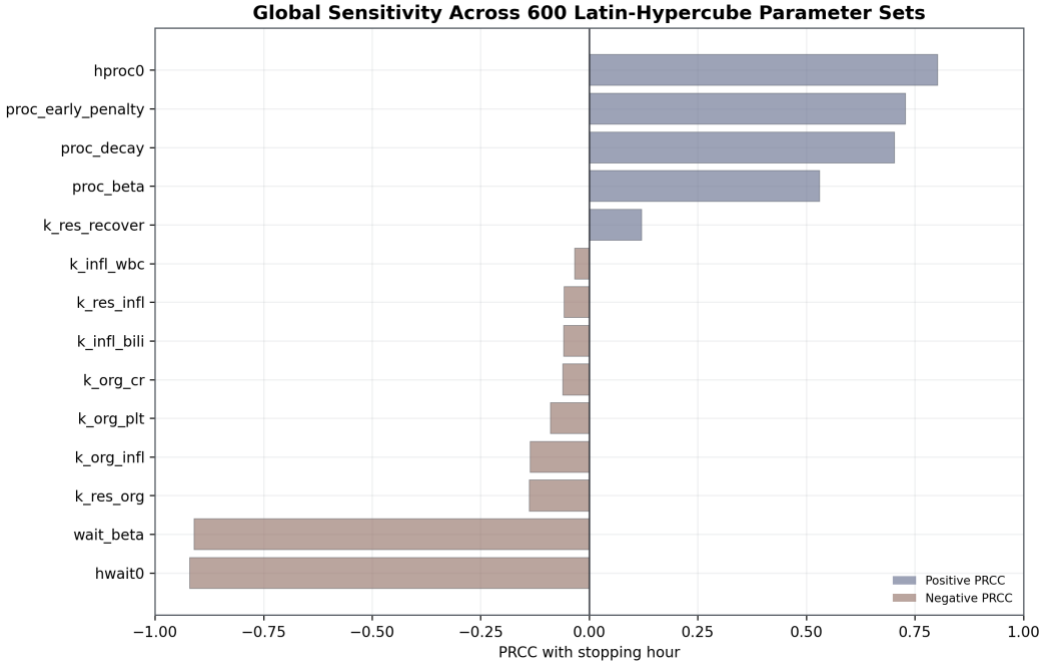


Figure 8. *Global PRCC sensitivity across 600 Latin-hypercube parameter sets.*

These associations are conditional on the selected parameter ranges and do not measure clinical causal effects. Local ±1% perturbations did not change the integer hour stopping decision for eight physiological parameters. In contrast, six decision cost parameters produced a one-hour shift for one perturbation direction, yielding reported finite difference elasticities of ±1.724. Because the endpoint is discretized to hourly decisions, these values characterize threshold crossings. The prominence of decision cost parameters highlights the need for empirical calibration before clinical interpretation.

3.8 Stochastic Initial Condition Uncertainty

In 1,000 stochastic perturbations of the five laboratory measurements and three latent initial states, the median candidate stopping hour was 28, with 2.5th and 97.5th percentiles of 19 and 40 hours (Figure 9). The distribution was concentrated around the reference decision of 29 hours but included earlier and later stopping decisions, showing dependence on the assumed starting profile. These percentiles describe uncertainty under the specified computational input distributions. Taken together with the global sensitivity results, they show that both initial state uncertainty and provisional decision cost assumptions materially affect the timing output. The findings support further calibration and validation rather than immediate use as patient-specific ERCP guidance.

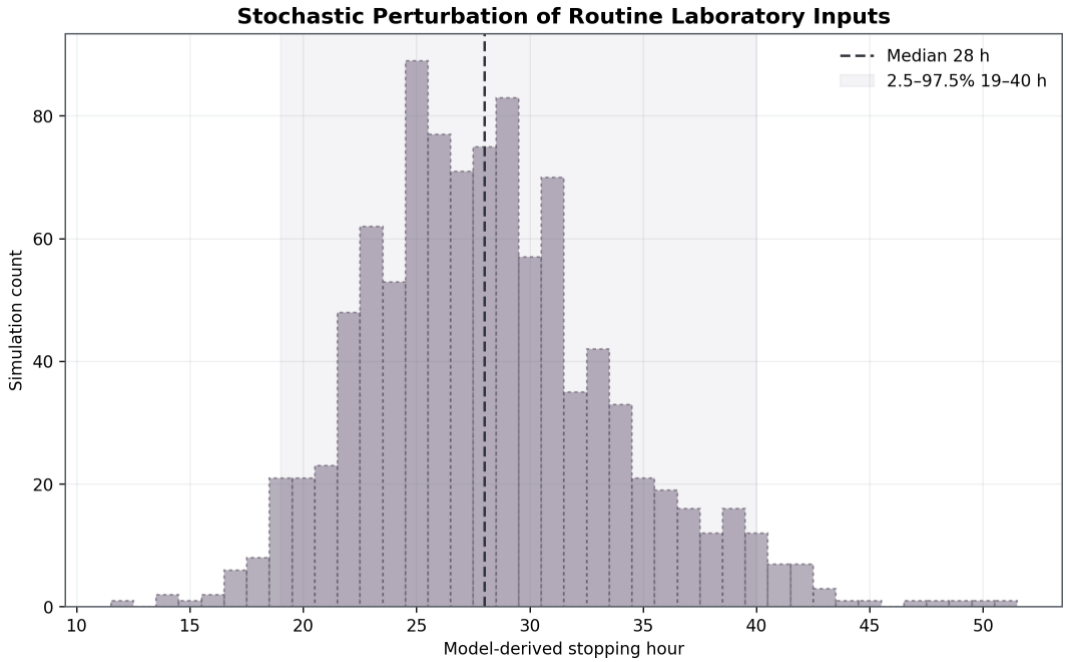


Figure 9. *Distribution of candidate stopping hours from 1,000 perturbed initial profiles. The vertical dashed line denotes the simulated median (28 hours), and the shaded band denotes the empirical 2.5^{th}-97.5^{th} percentile range (19-40 hours).*

4. Discussion

This study examined a theoretical waiting versus drainage decision in acute cholangitis using coupled laboratory and latent state ODEs, a four-state continuous time Markov chain, and a finite horizon Bellman recursion. In the reference simulation, bilirubin and creatinine increased, platelets and albumin declined, and the modeled stability reserve fell, although WBC decreased. At 24 hours, the deterioration index was 0.471 and reserve was 0.220. The first drainage favored Bellman decision occurred at 29 hours. The competing intensity functions explicitly represented the assumed cost of continued observation against an initially higher procedure related cost. This construction is consistent with the clinical

importance of severity assessment and timely biliary decompression discussed in TG18 and ASGE guidance, but the modeled cost functions and stopping hour were not estimated from clinical outcomes.

The timing decision varied with the prespecified laboratory and reserve profiles. Across eight scenarios, stopping hours ranged from 8 for the combined severe profile to 57 for the lower inflammatory profile, thrombocytopenic deterioration and emerging renal dysfunction yielded 16 and 21 hours, respectively. The 483-point bilirubin creatinine sweep similarly produced earlier stopping decisions at generally higher initial laboratory abnormalities, with outputs spanning 15-52 hours. These simulation shows the internal response of the specified decision rule. The 24-48-hour timing band used for comparison was contextual and was not imposed on the optimization. Published drainage timing studies and guidelines provide a rationale for studying timing, but cannot independently validate this simulated stopping surface.

The latent state analyses supplied complementary descriptions of the assumed deterioration process. In the phase plane, inflammatory burden and organ dysfunction increased along a curved trajectory. The CTMC shifted occupancy away from the initially dominant stable state toward organ dysfunction and intervention favored states. These outputs are consequences of the modeled coupling and transition intensities, and CTMC occupancy did not determine the Bellman action. The separately reported post drainage eigenvalues were all negative (-0.180 to -0.035 h^{-1}), which is compatible with local asymptotic stability only if they arise from the Jacobian at a genuine equilibrium of an explicitly specified recovery system. They do not establish global stability, clinical recovery after ERCP, or stability of the untreated equations.

Uncertainty analysis exposed an important dependence on provisional assumptions. Among 600 Latin hypercube parameter sets, stopping time was most strongly associated with baseline waiting intensity (PRCC -0.920), its slope (-0.910), and baseline procedure intensity (0.802), showing that decision cost specification dominated several physiological couplings. The hourly decision grid also limited interpretation of local ±1% elasticities. Separately, 1000 perturbed initial profiles yielded a median stopping time of 28 hours and a 2.5^{th}-97.5^{th} percentile range of 19-40 hours, these percentiles reflect assumed simulation inputs. Accordingly, the main contribution is a reproducible, falsifiable mathematical formulation that identifies quantities requiring longitudinal measurement and calibration. Before clinical application, future work must estimate state dynamics and decision costs from patient data, define the post drainage system, assess discrimination and calibration, and test whether model informed timing improves outcomes relative to established clinical practice.

5. Limitations

This study is a literature-parameterized mathematical proof-of-concept, not a clinically fitted or validated ERCP timing model. The reference profile and eight scenarios were illustrative. The ODE coefficients, laboratory transformation scales, deterioration index weights, Markov transition intensities, and Bellman cost parameters were provisional. Accordingly, the 29-hour reference stopping point, the 8-57-hour scenario range, and the bilirubin creatinine stopping surface describe behavior under the specified assumptions and cannot be interpreted as clinical thresholds or evidence of treatment benefit. In particular, the strong PRCC associations for waiting and procedure related cost parameters show that the decision is sensitive to assumptions that still require empirical calibration. The 1,000 run stochastic

experiment perturbed assumed initial inputs. Its 19-40-hour percentile interval is not a population prediction interval. The illustrative albumin threshold of 2.8 g/dL was based on an assumed lower limit of normal of 4.0 g/dL and would require laboratory-specific adaptation.

The pre-drainage model does not simulate changes following ERCP, and the Markov occupancy probabilities are theoretical rather than observed transitions, they do not drive the Bellman decision. The reported negative post drainage eigenvalues support a local stability interpretation only after an explicit recovery vector field and its equilibrium have been verified. Numerical clipping in the repeated simulation solver is also not a proof of global positivity or boundedness. The decision layer omits important determinants of actual care, including infection etiology, antimicrobial and resuscitation response, hemodynamic and respiratory status, altered biliary anatomy, procedural contraindications, and ERCP access. Neither clinical calibration, causal treatment effects, discrimination, nor external validation was performed. These limitations preclude patient level timing recommendations and require testing against serial clinical measurements and observed outcomes before any decision support use.

6. Conclusion

The coupled ODE, CTMC, and finite horizon stopping model provides a reproducible way to examine how assumed changes in routine laboratory markers, latent organ dysfunction, and stability reserve influence a theoretical waiting versus drainage decision in acute cholangitis. In the reference simulation, the first drainage favored decision occurred at 29 hours, with earlier or later decisions under alternative profiles, uncertainty analyses showd substantial dependence on provisional decision cost parameters and initial conditions. These results show internal computational behavior, not an optimal clinical ERCP time. The model may inform future registry design and hypothesis testing by identifying variables, transition assumptions, and costs that require estimation. Clinical usefulness remains contingent on explicit post intervention modeling, calibration, and independent validation alongside established severity assessment and specialist judgment.

References

Buxbaum, J. L., Buitrago, C., Lee, A., Elmunzer, B. J., Riaz, A., Ceppa, E. P., Al-Haddad, M., Amateau, S. K., Calderwood, A. H., Fishman, D. S., Fujii-Lau, L. L., Jamil, L. H., Jue, T. L., Kwon, R. S., Law, J. K., Lee, J. K., Naveed, M., Pawa, S., Sawhney, M. S., . . . Wani, S. (2021). ASGE guideline on the management of cholangitis. Gastrointestinal Endoscopy, 94(2), 207-221.e14. https://doi.org/10.1016/j.gie.2020.12.032

Choday S et al. Timing of intervention assessing early versus late endoscopic and percutaneous drainage in acute cholangitis. World J Gastrointest Endosc. 2026.

Comoglu M et al. Effect of urgent ERCP on clinical outcomes in acute cholangitis and gallstone pancreatitis. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12380036/

Du, L., Cen, M., Zheng, X., Luo, L., Siddiqui, A., & Kim, J. J. (2020). Timing of performing endoscopic retrograde cholangiopancreatography and inpatient mortality in acute cholangitis: A systematic review and meta-analysis. Clinical and Translational Gastroenterology, 11(3), e00158. https://doi.org/10.14309/ctg.0000000000000158

Jagtap, N., Rughwani, H., Talukdar, R., Chavan, D., Memon, S. F., Asif, S., Kulkarni, A. V., Kalapala, R., Ramchandani, M., Lakhtakia, S., Darisetty, S., Rao, G. V., Tandan, M., Bruno, M. J., & Reddy, N. D. (2026). Urgent versus early ERCP in mild-to-moderate acute cholangitis: A randomised controlled trial. Gut. Advance online publication. https://doi.org/10.1136/gutjnl-2025-337279

Kiriyama, S., et al. (2018). Tokyo Guidelines 2018: Diagnostic criteria and severity grading of acute cholangitis (with videos). Journal of Hepato-Biliary-Pancreatic Sciences, 25(1), 17-30. https://doi.org/10.1002/jhbp.512

Kongsakon R et al. Urgent ERCP within 24 h for moderate-to-severe cholangitis due to CBD stones propensity score matched study. Surg Endosc. 2025. https://pubmed.ncbi.nlm.nih.gov/41023220/

Lyu, Y., et al. (2022). Impact of timing of ERCP in acute cholangitis: A systematic review and meta-analysis. World Journal of Gastroenterology. https://pmc.ncbi.nlm.nih.gov/articles/PMC9719833/

Mayumi, T., et al. (2018). Tokyo Guidelines 2018: Management bundles for acute cholangitis and cholecystitis. Journal of Hepato-Biliary-Pancreatic Sciences. https://doi.org/10.1002/jhbp.519

Miura, F., et al. (2018). Tokyo Guidelines 2018: Initial management of acute biliary infection and flowchart for acute cholangitis. Journal of Hepato-Biliary-Pancreatic Sciences. https://doi.org/10.1002/jhbp.509

Mohan R et al. Validation of TG07 and TG13/TG18 criteria and predictors of mortality in patients over 80 years. Clin Exp Hepatol. 2021.

Mukai, S., et al. (2018). Indications and techniques of biliary drainage for acute cholangitis in updated Tokyo Guidelines 2018. Journal of Hepato-Biliary-Pancreatic Sciences. https://doi.org/10.1002/jhbp.496

Tan M et al. Association between early ERCP and mortality in acute cholangitis. Gastrointest Endosc. 2018. https://pubmed.ncbi.nlm.nih.gov/28433613/

Wada K et al. Diagnostic criteria and severity assessment of acute cholangitis Tokyo Guidelines. J Hepatobiliary Pancreat Surg. 2007. https://pmc.ncbi.nlm.nih.gov/articles/PMC2784515/

Zhu Y et al. Association of timing of biliary drainage with clinical outcomes in severe acute cholangitis. Int J Gen Med. 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC8254098/